



Spring, 2026

MCT344-342: Industrial Robotics

Sheet 02: Rigid Motions

- 1) Consider the following sequence of rotations:
 1. Rotate by ϕ about the world **x-axis**.
 2. Rotate by θ about the current **z-axis**.
 3. Rotate by ψ about the world **y-axis**.
 Write the matrix product that will give the resulting rotation matrix (**do not perform the matrix multiplication**).

- 2) Consider the following sequence of rotations:
 1. Rotate by ϕ about the world **x-axis**.
 2. Rotate by θ about the current **z-axis**.
 3. Rotate by ψ about the current **x-axis**.
 4. Rotate by α about the world **z-axis**.
 Write the matrix product that will give the resulting rotation matrix (**do not perform the matrix multiplication**).

- 3) If the coordinate frame **01x1y1z1** is obtained from the coordinate frame **00x0y0z0** by a rotation of $\frac{\pi}{2}$ about the **x-axis** followed by a rotation of $\frac{\pi}{2}$ about the fixed **y-axis**, find the rotation matrix **R** representing the composite transformation. Sketch the initial and final frames.

- 4) Suppose that three coordinate frames **01x1y1z1**, **02x2y2z2**, and **03x3y3z3** are given, and suppose:

$$R_2^1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}, R_3^1 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Find the matrix **R**₃²

- 5) Find the rotation matrix corresponding to the Euler angles $\phi = \frac{\pi}{2}$, $\theta = 0$, $\psi = \frac{\pi}{4}$, What is the direction of the **x1** axis relative to the base frame?

- 6) Compute the homogeneous transformation representing a translation of 3 units along the **x-axis** followed by a rotation of $\frac{\pi}{2}$ about the current **z-axis** followed by a translation of 1 unit along the fixed **y-axis**. Sketch the frame. What are the coordinates of the origin **O1** with respect to the original frame in each case?

- 7) Consider the diagram of **Figure 1**. Find the homogeneous transformations H_1^0, H_2^0, H_2^1 representing the transformations among the three frames shown. Show that $H_2^0 = H_1^0, H_2^1$.

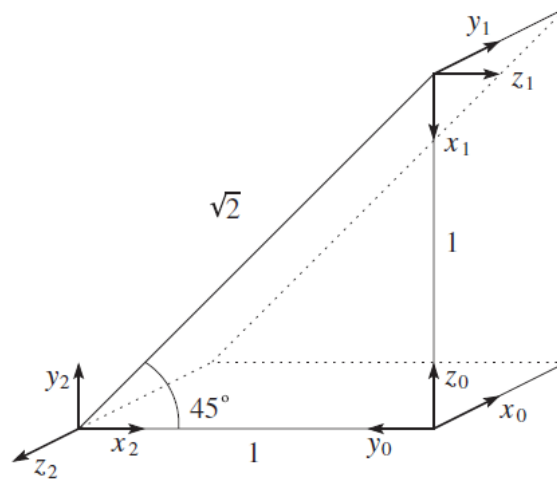


Figure 1

- 8) Consider the diagram of Figure 2. A robot is set up 1 meter from a table. The tabletop is 1 meter high and 1 meter square. A frame $01x_1y_1z_1$ is fixed to the edge of the table as shown. A cube measuring 20 cm on a side is placed in the center of the table with frame $02x_2y_2z_2$ established at the center of the cube as shown. A camera is situated directly above the center of the block 2 meters above the tabletop with frame $03x_3y_3z_3$ attached as shown. Find the homogeneous transformations relating each of these frames to the base frame $00x_0y_0z_0$. Find the homogeneous transformation relating the frame $02x_2y_2z_2$ to the camera frame $03x_3y_3z_3$.

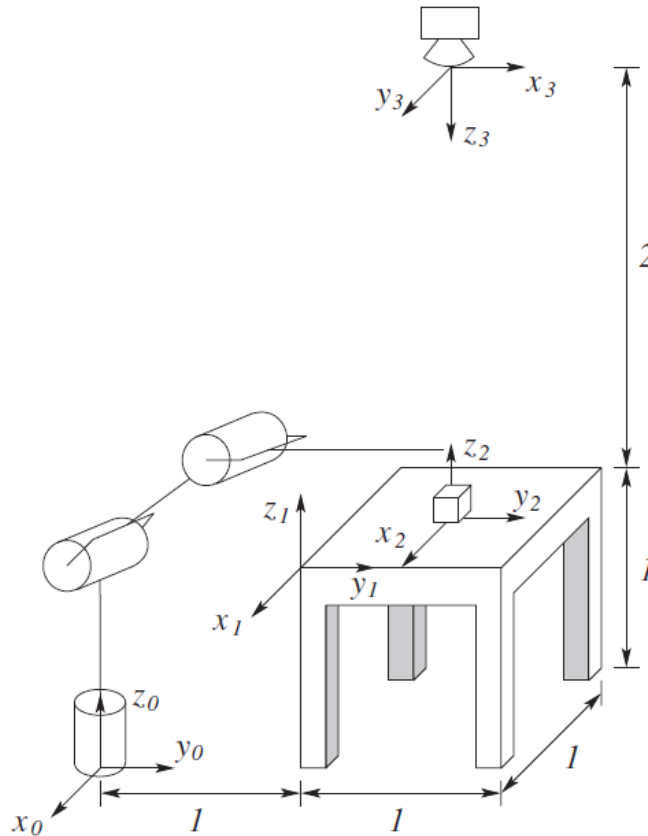


Figure2

- 9) In ***Problem 8***, suppose that, after the camera is calibrated, it is rotated 90° about z_3 . Recompute the above coordinate transformations.
- 10) If the block on the table is rotated 90° about z_2 and moved so that its center has coordinates $[0, 0.8, 0.1]^T$ relative to the frame $o_1x_1y_1z_1$, compute the homogeneous transformation relating the block frame to the camera frame; the block frame to the base frame.